EXPERIMENTAL DESIGN HW 1

Z1151203 張芯君

1. (a) $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{64}} = 1$

(b) 95% $\Rightarrow M \pm 1.96\sigma_{\bar{x}} \Rightarrow \pm 1.96 \times 1 = \pm 1.96$

(c) $\bar{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} = \bar{X} \pm 1.96\,\sigma_{\bar{x}} = 105 \pm 1.96 \Rightarrow (103.04, 106.96)$

2. (a) $\frac{320}{400} = 80\%$

(b) $\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.8 \pm 1.96 \sqrt{\frac{0.8 \times 0.2}{400}} \Rightarrow (0.7608, 0.8392)$

(c) We have 95% confident that the proportion of components meeting the required quality standards is between 76.08% and 83.92%.

3. (a) $\bar{X} = \frac{800}{8} = 100$,

$S = \sqrt{\frac{\Sigma(X_i - \bar{X})^2}{n-1}} = \sqrt{\frac{1}{7}(4+9+1+1+9+4)} = 2$

(b) $\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} = 100 \pm 2.365 \frac{2}{\sqrt{8}} \fallingdotseq 100 \pm 1.6723 \Rightarrow (98.3277, 101.6723)$

(c) $df = 7$

4. (a) $H_0: M = 500 \quad H_1: M < 500$

(b) one sample Z test is appropiate. $Z = \frac{\bar{X} - M}{\frac{\sigma}{\sqrt{n}}}$

reasons why suitable: ① σ 已知 ② random sample $n = 64 > 30$ ③ 檢驗的是單一母體的平均數

(c) $Z = \frac{498.5 - 500}{8/\sqrt{64}} = -1.5$

p-value $= P(Z \leq -1.50) = 0.0668$

(d) p-value $= 0.0668 > \alpha = 0.05$

$\Rightarrow$ Do not reject H_0, since there is insufficient evidence to conclude that the machine dispenses fewer than 500ml per bottle at the significance level $\alpha = 0.05$.

5. (a) $H_0: M = 40 \quad H_1: M \neq 40$

(b) t distribution, $df = 36 - 1 = 35$

(c) $t = \frac{\bar{X} - M_0}{\frac{S}{\sqrt{n}}} = \frac{39.5 - 40}{6/\sqrt{36}} = -2.5$

(d) $t_{\frac{\alpha}{2}, n-1} = t_{0.025, 35} = 2.030 \Rightarrow$ If $t < -2.03$ or $t > 2.03$, then reject H_0

Since $t = -2.5 < -2.03$ reject H_0, there is sufficient evidence to conclude that the population mean inspection time differs from 40 min at the significance level $\alpha = 0.05$.